

AmropaliNet: Detect early. Spray less. Waste less.

AI mango doctor for the Amrapali farmers of Bangladesh - protecting orchards, soil, water and bees

Problem Statement

272 words

Amrapali is Bangladesh's favourite everyday mango and the main source of income for thousands of farming families in Rajshahi, Chapainawabganj, Naogaon and the hill districts. Every season it is attacked by six common diseases: anthracnose, bacterial canker, powdery mildew, scab, sooty mould and stem-end rot.

Most farmers cannot tell these diseases apart, and an agricultural officer is rarely at hand when the first spots appear. The usual answer is blanket, calendar spraying: the whole orchard is sprayed with several fungicides and insecticides "just in case", again and again from flowering to harvest.

This turns a farming problem into an environmental one. Surplus chemicals wash into soil, ponds and groundwater, harming aquatic life and village drinking water (SDG 6). Spraying at flowering kills the bees and other pollinators that mango trees need to set fruit, and repeated doses breed resistant pathogens (SDG 15). Farm workers and their families breathe and handle these chemicals, often without protection (SDG 3).

The second problem is waste. When disease is noticed late, it has already spread, and infected fruit rots on the tree, in storage or in transport. Every discarded mango wastes the water, land, fertiliser, fuel and labour used to grow it, and rotting fruit releases greenhouse gases (SDG 12, SDG 13). Lost harvests and wasted chemical costs push small farmers towards poverty and shrink the local food supply (SDG 1, SDG 2).

ROOT CAUSE

The root cause is one missing piece of knowledge at the right moment: which disease is this, and what is the smallest treatment that works? Without it, farmers overspray, lose fruit and damage the ecosystem their orchards depend on - season after season.

Environmental harm and the SDG targets it violates

Environmental problem	SDG	Target that is violated
Chemical runoff into soil, ponds and groundwater	SDG 6	6.3 Improve water quality by reducing pollution and hazardous chemicals
Blanket, calendar spraying of agro-chemicals	SDG 12	12.4 Sound management of chemicals; less release to air, water and soil
Pollinators killed, resistant pathogens bred	SDG 15	15.5 Halt the loss of biodiversity
Farm families exposed to pesticides	SDG 3	3.9 Fewer illnesses from hazardous chemicals and pollution
Diseased fruit rots before it is eaten	SDG 12	12.3 Reduce food losses along production and supply chains
Wasted water, land, fuel and emissions	SDG 13	13 Take urgent action to combat climate change and its impacts
Lost harvests and income, less local food	SDG 2	2.4 / 1.5 Sustainable, resilient food production; resilience of the poor

Where: Rajshahi, Chapainawabganj, Naogaon and the hill districts

Who is affected: small mango farmers, farm workers, consumers, bees and orchard wildlife

AmropaliNet: the right treatment, only where needed

One photo, the right diagnosis, less chemical and less waste - a healthier environment

Solution

297 words

AmropaliNet is a free Bangla and English AI mango doctor that replaces "just in case" spraying with precise, evidence-based treatment.

We built a dataset of 3,500 real Amrapali images - 500 for each of 7 classes (six diseases and healthy) - split 70/15/15 for training, validation and testing. On it we trained our proposed model, AA-ENet: an EfficientNet-B0 backbone with CBAM attention, which focuses on the lesion instead of the background, and a Transformer encoder, which links spots across the whole fruit. AA-ENet reaches 96.8% test accuracy (F1 0.968, 96.6% in cross-validation) with only 4.57 million parameters: the same accuracy as Xception with 4.6 times fewer parameters, and higher than VGG16, which is 29 times larger.

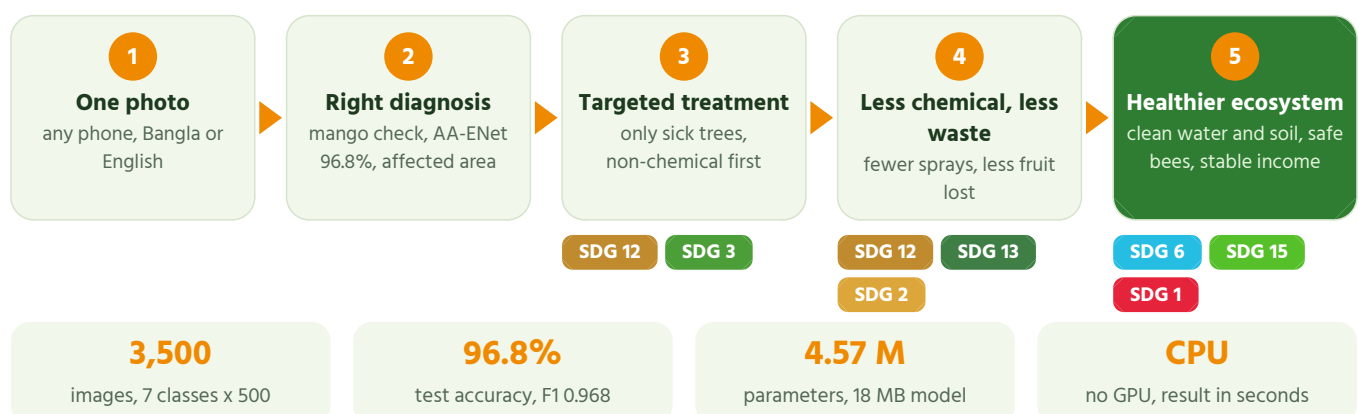
A farmer photographs a mango with any phone. A two-step mango check (CLIP plus a confidence gate) rejects other photos, AA-ENet names the disease in seconds, and Grad-CAM outlines the affected area, so the farmer sees the evidence before acting.

How each diagnosis protects the environment:

- No disease, no spray: a healthy result ends the calendar spray for that tree (SDG 12.4).
- No wrong spray: sooty mould grows on honeydew from sap-sucking insects, so fungicide alone only pollutes; the app advises controlling the insects, with neem oil first because it is gentler on bees (SDG 6.3, SDG 15.5).
- Only sick trees, non-chemical steps first: pruning, orchard hygiene and hot-water treatment after harvest reduce the chemicals farm families handle (SDG 3.9).
- Early detection stops spread, saving fruit and the water, land and emissions behind it (SDG 12.3, SDG 13).
- Green AI: the 18 MB model runs on an ordinary CPU, with no GPU servers and little energy per diagnosis.

The engine is the open Python library mango-disease-ai, so extension services can bring it to every orchard. Detect early. Spray less. Waste less.

How everything connects: from one photo to the SDGs



Same accuracy, far smaller model = less energy per diagnosis

AA-ENet (ours)	● 4.57 M parameters 96.8%
Xception	20.82 M parameters 96.8%
VGG16	134.29 M parameters 95.6%

SDGs AmropaliNet advances:

